

Research Article

Water-Soluble Inorganic Anions in Fine Particulate Matter (PM_{2.5}) in Dhaka, Bangladesh: Source Apportionment

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Bangladesh is one of the most populated countries in the world. More than 200 air quality index values were recorded in January and February since 2017, indicating that the air quality was unhealthy to very unhealthy. In this work, water-soluble inorganic anions (WSIAs) in fine particulate matter (PM_{2.5}) in Dhaka, the capital of Bangladesh with a population of around 21 million, were evaluated. Accordingly, PM_{2.5} was collected using a quartz filter coupled with an active air sampler set in Dhaka University campus, Bangladesh, during February 2023, to analyze WSIAs. The 24-h average PM_{2.5} mass, $156 \pm 44.8 \mu\text{g}/\text{m}^3$, was almost nine times higher than the WHO limit values of $15.0 \mu\text{g}/\text{m}^3$ and about three times higher than the Bangladesh Department of Environment limits of $65.0 \mu\text{g}/\text{m}^3$. Fluoride, chloride, bromide, nitrite, nitrate, phosphate, and sulfate were among the water-soluble ions that were determined using ion chromatography. While bromide and nitrite were found to be below the detection limit, the mean concentrations of fluoride, chloride, nitrate, phosphate, and sulfate were determined to be 0.086 ± 0.010 , 2.27 ± 0.332 , 10.2 ± 1.29 , 2.23 ± 0.424 , and $11.0 \pm 1.45 \mu\text{g}/\text{m}^3$, respectively. The highest levels of PM_{2.5} and the relevant anions were determined to be present during February 9–10, 2023. Sulfate accounted for almost 43% of the anions in the overall PM_{2.5}. According to the results of the statistical analysis, the high concentrations of sulfate and nitrate in PM_{2.5} in Dhaka, Bangladesh, were primarily caused by sulfur-containing fossil fuels, especially sulfur-containing diesel, coal, and unfit automobiles, while the main sources of fluoride and chloride were fire retardants, burning of textile, and municipal solid wastes. All of these sources have the ability to produce PM_{2.5}.

Keywords: air pollution; fossil fuels; particulate matter; unfit automobiles; very unhealthy; water-soluble anions

1. Introduction

Dhaka is the capital city of Bangladesh, with a population of about 21 million [1–4]. In addition to being the largest city in the whole country, it is also the ninth-largest metropolis worldwide. Due to these factors, in addition to being the center of the country's economic life, Dhaka experiences fairly high levels of pollution each year, with

only brief respite periods that, despite being lower, still have some relatively high pollution readings, making the city's air unhealthy to breathe year-round [5–8]. As per the 2018 rankings issued by IQAir, a Swiss-based air quality technology company, Bangladesh is the most polluted country on the earth [7–9].

The air quality index (AQI), a daily air quality reporting index, lets people know how clean or dirty the air in

a particular city is and whether any potential health impacts need their attention. Dhaka, which frequently tops lists of the world's most polluted cities, has experienced record-breaking air pollution in 2018 [7–9]. The IQAir 2021 World Air Quality Report states that Bangladesh's average $\text{PM}_{2.5}$ concentration was $76.9 \mu\text{g}/\text{m}^3$, which is 15 times higher than the WHO recommendation ($15 \mu\text{g}/\text{m}^3$) [7–10]. Dhaka became the second most polluted regional capital city globally, behind only New Delhi in neighboring India, with an annual average $\text{PM}_{2.5}$ concentration of $78.1 \mu\text{g}/\text{m}^3$ [8]. According to Rahman et al. (2021), the concentration of $\text{PM}_{2.5}$ ranged from 5.27 to $105 \mu\text{g}/\text{m}^3$, with an annual average (January–December 2016) of $32.7 \pm 20.3 \mu\text{g}/\text{m}^3$ [11]. In addition, they found that $\text{PM}_{2.5}$ had a high sulfur concentration ($1.2 \pm 0.8 \mu\text{g}/\text{m}^3$). A different study conducted in central China from January 13 to January 24, 2018, showed that the average daily concentration of $\text{PM}_{2.5}$ was $92.9 \mu\text{g}/\text{m}^3$ [12]. They reported that the water-soluble inorganic ions (WSIIs), which were mostly composed of ammonium, sulfate, and nitrate, accounted for 57.3% of the total $\text{PM}_{2.5}$ concentration over the course of the study period.

According to the USEPA, the Centre for Atmospheric Pollution Studies (CAPS) at Stamford University, Bangladesh, noticed that the number of days with hazardous air quality counted was five in 2017, six in 2018, six in 2019, seven in 2020, twelve in 2021, and five in 2022 [9]. The number of days with “very unhealthy” air quality counted was 64 in 2017, 79 in 2018, 60 in 2019, 69 in 2020, 87 in 2021, and 77 in 2022. The AQI in Bangladesh is based on five pollutants: particulate matter (PM_{10} and $\text{PM}_{2.5}$), NO_x , CO , SO_x , and ozone. It should be noted that all sorts of pollutants, including persistent organic pollutants (POPs), heavy metals, different anions, and organometallic compounds (e.g., organotin and organomercury), are transported by atmospheric fine particulate matter, particularly $\text{PM}_{2.5}$. Typically, the winter months are when the air quality deteriorates and the monsoon months are when it improves, but it typically stays poor.

The average life expectancy of a Bangladeshi resident has decreased by 2.1 years since 1998 due to a 39% increase in the country's yearly average particulate pollution [8]. It has been reported that during the months of January and February, a remarkably high number of patients in Dhaka city were receiving diagnoses for poor respiratory diseases on a daily basis [13]. The fact that most of these patients have no previous record of severe illness and that they primarily depend on outside air exposure to fulfill their work obligations suggests that the illness may be linked to air pollution. According to reports, air pollution caused 24,000 premature deaths in Dhaka between 2005 and 2018 [14]. Among the 46 cities investigated, that was the highest. Dhaka has experienced the highest rise in the number of premature deaths among the top 15 most polluted cities in the world, the majority of which are in Asia [14].

Particulate matter (PM) itself causes breathing problems like asthma, and PM linked to other pollutants such as POPs, heavy metals, and anions must have detrimental consequences on health like cancer [4, 6, 15, 16]. Aerosols in the atmosphere, mixture of particles (solid or liquid), have sizes

between 10^{-9} and 10^{-4} m and can be either solid, liquid, or both. Aerosol particles vary widely in terms of their size, origin, chemical makeup, quantity, dispersion, and time spent in the atmosphere [17, 18]. The airborne particles are produced by erosive processes, sea spray, and volcanic eruptions, among other natural occurrences [19–21]. Water-soluble ions are one of the major components of atmospheric aerosols, particularly $\text{PM}_{2.5}$. They may make up to 60%–70% of the majority of suspended PM. The atmospheric aerosol's water-soluble component is hygroscopic and is made up of several important materials [17, 18]. Along with changing their optical characteristics, water-soluble aerosols make dangerous organic compounds more soluble by serving as surface active reagents, which increase their toxicity to human health. The distribution and concentration of water-soluble ions can, therefore, reveal source contributions, atmospheric chemical alterations, and their possible negative impacts on the environment and human health.

In Dhaka, many attempts have been made to characterize and identify the sources of atmospheric aerosols, including $\text{PM}_{2.5}$ and PM_{10} [22–27]. So far, Salam and co-workers have looked into the quantities of water-soluble ions present in the atmospheric aerosols during day and night time [19, 27]. With the exception of potassium, nitrate, and bromide, they found that the concentrations of calcium, sodium, phosphate, chloride, and sulfate were significantly higher at night than those during the daytime. Despite this, the city has consistently been rated among the worst in the world for air quality over the past few years. The levels of SO_x and NO_x in the air have been substantially raised by the vastly increased growth of the brickfields surrounding Dhaka City and the running of a massive number of unfit vehicles. In addition, the levels of SO_x and NO_x in the air in Dhaka are integrated by the open combustion of municipal solid wastes. In this study, $\text{PM}_{2.5}$ was collected using an active air sampler during February 2023 to investigate water-soluble inorganic anions (WSIAs), such as fluoride, chloride, bromide, nitrite, nitrate, phosphate, and sulfate and their potential sources.

2. Methods and Experimental

2.1. Study Area. An active air sampler (Model: TH-150H) was set up on the roof (about 45 m above the ground) of the Mukarram Hussain Khundker Biggyan Bhaban, Dhaka University, Bangladesh ($23^{\circ}43'42.70''$ N, $90^{\circ}23'54.90''$ E), to collect $\text{PM}_{2.5}$ (Figure 1). The study site was not surrounded by any elevated structures or sources of industrial air pollution; therefore, it accurately represented the pollution patterns in Dhaka.

2.2. Sample Collection. As it is well-known, January and February of each year are the polluted months in Dhaka [9]. Thus, the sample of $\text{PM}_{2.5}$ was collected on quartz filters (PALL Company, USA) with a 90 mm diameter from February 2, 2023, to February 27, 2023. Each sampling session lasted for 24 h (from 8:00 a.m. to the following 8:00

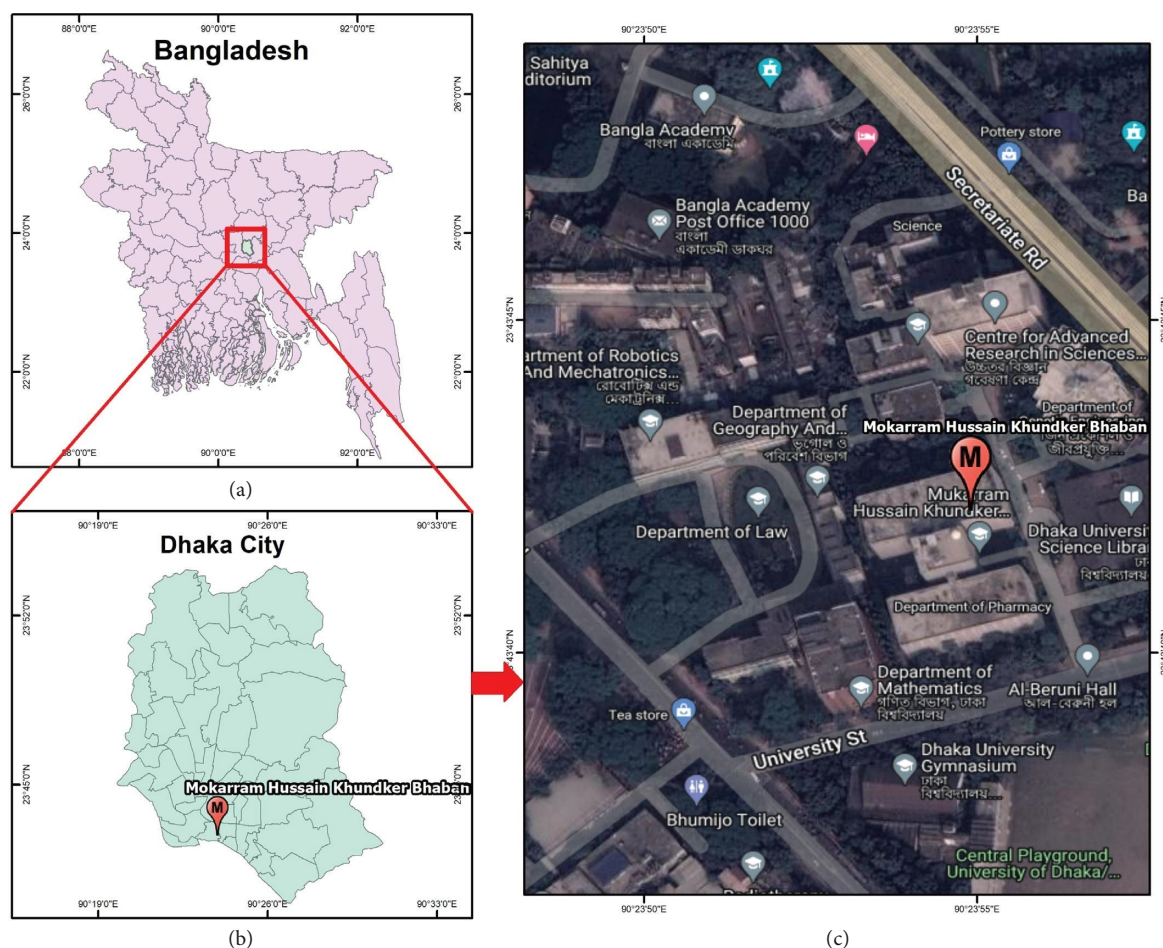


FIGURE 1: Map of Bangladesh (a), Dhaka city (b), and sampling site (c).

a.m.), and we employed a median volume sampler together with $PM_{2.5}$ cutting equipment (Model: TH-150H, China) at a flow rate of 70 L/min. There were 10 valid samples in all. A field blank (one filter) was treated in the same way as loaded filters for the purposes of blank correction. The filters were baked at 250°C for 12 h before sampling to eliminate volatile compounds and other contaminants. Prior to sampling, the filters were also weighed under controlled temperature and humidity conditions (25°C, 50% RH). Following weight analysis, the filters were sealed and kept in a refrigerator for ion analysis. The weight increase of the quartz filter was compared using PTFE filter (Merck KGaA, Darmstadt, Germany; pore size: 2.0 μm and diameter: 47 mm) by moisture in maintaining ambient conditions for 24 h throughout the sampling period. It should be pointed out that neither filter showed any weight gain. This could be a result of the low relative humidity (approximate 49%) during January-February in Dhaka. Supporting Table S1 displays an illustration of a description. Supporting Figure S1 displays the quartz filters both before and after use.

2.3. Chemical Analysis. The quartz filter, which had a 90 mm diameter, was divided into four pieces, and only a portion of it was treated in 20 mL of deionized water using an automated shaker and sonication for 30 min. After being left to

remain still for 60 min, the supernatant was collected for analysis. The detection and measurement of the water-soluble anions in $PM_{2.5}$ was performed using an ion chromatography (Model: S7150, Sykam, Germany). ION A06 (4.0 $\times$ 250 mm) column was used to analyze the WSIA, particularly F^- , Cl^- , Br^- , NO_2^- , NO_3^- , PO_4^{3-} and PO_4^{2-} , under the following conditions: 1.7 mM $NaHCO_3$ and 1.8 mM Na_2CO_3 solution was used as eluent with a flow rate of 1.0 mL/min; limits of detection (LODs) were recorded when $S/N \geq 3$; gourd column: AGC-3 (4.6 $\times$ 10 mm); injection volume 20 μL [27]. Standard solutions with a concentration range of 1–10 ppm for each anion were prepared from the appropriate certified commercial solution (MERCK Suprapur solutions, Hong Kong, China) upon dilution using deionized water in order to analyze the anions. These solutions were used to construct calibration curves with a coefficient correlation of at least $R^2 > 0.995$. The values of LOD of F^- , Cl^- , NO_2^- , Br^- , NO_3^- , PO_4^{3-} , and PO_4^{2-} were 0.033, 0.049, 0.078, 0.114, 0.098, 0.332, and 0.073 $\mu g/m^3$, respectively. Concentrations of Br^- and NO_2^- were below LODs.

2.4. AQI Values for the Months of January and February During 2017–2023. A bar diagram for the AQI values calculated by the US Embassy air quality monitoring center in

Dhaka, Bangladesh, has been created for the months January and February during 2017–2023. The daily $\text{PM}_{2.5}$ ($\mu\text{g}/\text{m}^3$) for the 24-h concentration average from midnight to midnight (local standard time) was used to calculate the AQI values [9].

2.5. Meteorological Data and Data Processing. In the winter months of January and February, the wind in Dhaka city primarily flows from north to south [22]. Over time, this wind pattern shifts, eventually blowing from south to north during the premonsoon (July–August) and the monsoon (August–September) [7]. The Bangladesh Meteorological Data Network (<https://www.iqair.com/bangladesh/dhaka>) provided the meteorological and air quality data for the sample period (February 2, 2023–February 27, 2018). The monitoring station is the Mukarram Building at Dhaka University, which is also the global station [7] (Figure 1).

2.6. Statistical Analysis. The Pearson's correlation and principal component analysis (PCA) were computed using MS-Excel (XLSTAT) to identify linear correlation and potential sources of the variables such as WSIA in $\text{PM}_{2.5}$. PCA is a multivariate statistical analysis method that is based on the variable correlation matrix. It is used to investigate and simplify large, complex datasets so that qualities may be identified.

3. Results and Discussion

3.1. Air Quality Status in Dhaka. Figures 2(a) and 2(b) correspondingly exhibit the AQI values from 2017 to 2023 in Dhaka and the groups of health risks related to the AQI values [9]. Figure 2(a) shows that the values of AQI during January were somewhat higher than those during February. According to the findings, the air quality over the previous six to seven years ranged from bad to very harmful (Figure 2(b)). The ambient $\text{PM}_{2.5}$ has made an important contribution to the overall AQI values. In order to understand how humans are affected by chemical pollutants through transportation, measurement of WSIA in $\text{PM}_{2.5}$ is crucial. The 24-h average $\text{PM}_{2.5}$ mass, $156 \pm 44.8 \mu\text{g}/\text{m}^3$ (average of day and night) (Table 1), was almost 10 times higher than the WHO limit values of $15 \mu\text{g}/\text{m}^3$ [10] and about two and half times higher than the Bangladesh Department of Environment limits of $65.0 \mu\text{g}/\text{m}^3$ [27]. The mean value of $\text{PM}_{2.5}$ was found to be $191 \mu\text{g}/\text{m}^3$ in the samples collected between January 2013 and February 2013 (Table 2) [27].

Figure 3 displays the daily average concentrations of $\text{PM}_{2.5}$ and WSIA during the sample period, and the point-line diagram illustrates the simultaneous AQI, which is frequently used to assess air quality thoroughly [7]. During the 20-day study period, the AQI values were over 150 for 11 days, indicating unhealthy conditions, and over 200 for nine days, indicating very unhealthy conditions (Figure 3) [7].

$\text{PM}_{2.5}$ was found to be the main atmospheric pollutant during this time. The findings indicate that the daily average mass concentration of $\text{PM}_{2.5}$ throughout the sampling

period was determined to be $156 \pm 44.8 \mu\text{g}/\text{m}^3$, with a range of $107\text{--}221 \mu\text{g}/\text{m}^3$. However, in accordance with National Ambient Air Quality Standards, the allowable concentration of $\text{PM}_{2.5}$ for a 24-h averaging period was set at $65 \mu\text{g}/\text{m}^3$ in Bangladesh [35]. The results show that there was severely poor air quality during every day of the sample period.

3.2. Concentrations of WSIA in $\text{PM}_{2.5}$

3.2.1. Fluoride. The concentration of fluoride varied from 0.071 to $0.104 \mu\text{g}/\text{m}^3$, with a mean of $0.086 \pm 0.010 \mu\text{g}/\text{m}^3$. Fluoride was determined to have a maximum concentration during February 9–10, 2023 ($0.104 \mu\text{g}/\text{m}^3$) and a minimum concentration during February 2–3, 2023 ($0.071 \mu\text{g}/\text{m}^3$) (Table 1). However, high concentration of fluoride in $\text{PM}_{2.5}$ was reported by Salam et al., which was to be $0.13\text{--}0.35 \mu\text{g}/\text{m}^3$ [22]. Figure 3 suggests that the air quality gets quite bad in January and starts to get a little better in February. As a result, we observed that fluoride concentration peaked at the start of February and then decreased very slightly. Backward air trajectory data showed that the air direction altered very little from north to south throughout the sample period (Feb 2, 2023–Feb 27, 2023). This could be the cause of the consistency in fluoride concentration in the sampling period. Fluorinated compounds, particularly per- and polyfluoroalkyl substances (PFASs), are used in a variety of industries, including the production of rubber, plastics, and textiles, which contributes to the presence of fluoride in nature [36, 37].

Bangladesh is host to a huge number of textile industries, and PFASs are sometimes employed there to add properties like water and dirt resistance. Although there is limited information on PFASs in Bangladesh, textile waste emissions to the atmosphere and discharge into rivers and other water bodies could pose a serious threat to environment and human health. As a result, we investigated the detection and quantification of PFASs using a passive sampler made of a polyethylene (PE) sheet in Dhaka, and we found considerable levels of these fluorinated compounds in both the air and the water [37]. These results suggest that emerging contaminants, particularly PFASs, have not only contaminated the aquatic and biotic environment but are also causing pollution in the atmosphere of Bangladesh [5, 6, 37].

3.2.2. Chloride. The mean chloride concentration was found to be $2.27 \pm 0.332 \mu\text{g}/\text{m}^3$, with a range of $1.62\text{--}2.72 \mu\text{g}/\text{m}^3$ (Table 1). Similar to fluoride, the maximum concentration was found during February 9–10, 2023 ($2.72 \mu\text{g}/\text{m}^3$), while the lowest was observed during February 24–25, 2023 ($1.62 \mu\text{g}/\text{m}^3$). Salam and coworkers also looked into the concentration of chloride in $\text{PM}_{2.5}$ in Dhaka [22, 27]. They found that the concentrations of chloride were $1.23\text{--}3.08$ and $1.19 \pm 2.03 \mu\text{g}/\text{m}^3$ during March–April 2001 and January–February 2013, respectively [22, 27] (Table 2). Due to the considerable population expansion in the city of Dhaka, the huge amounts of municipal solid waste that were burned may have had high concentrations of bleaching powder [5, 6], which also contributed to the heightened

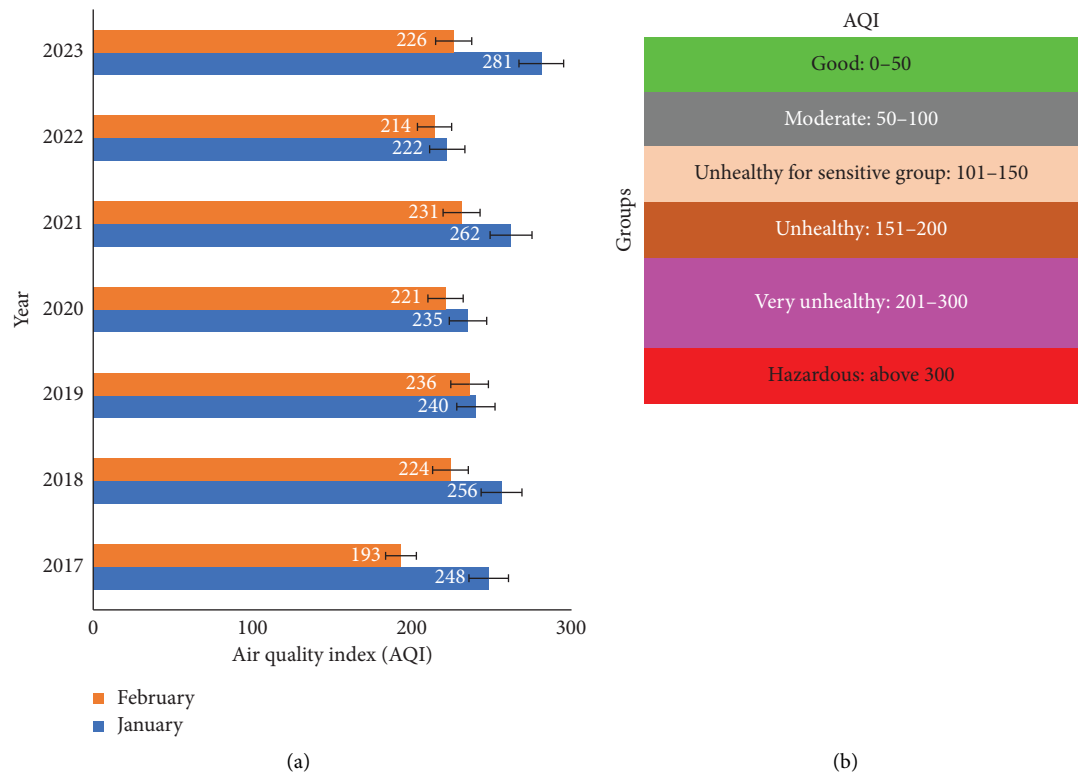


FIGURE 2: The AQI values from 2017 to 2023 in Dhaka (a) and the groups of health risks related to the AQI values (b) [9].

TABLE 1: Concentrations ($\mu\text{g}/\text{m}^3$) of water-soluble inorganic anions in $\text{PM}_{2.5}$ collected during February 02, 2023–February 27, 2023, Dhaka, Bangladesh.

Date	Fluoride	Chloride	Nitrate	Phosphate	Sulfate	TWSIA	PM _{2.5}
Feb 2, 2023–Feb 3, 2023	0.076	2.02	9.25	1.92	10.1	23.4	119
Feb 4, 2023–Feb 5, 2023	0.094	2.62	12.0	2.73	13.0	30.4	204
Feb 6, 2023–Feb 7, 2023	0.086	2.32	10.5	2.32	11.5	26.7	185
Feb 9, 2023–Feb 10, 2023	0.104	2.72	13.0	2.93	14.0	32.7	221
Feb 11, 2023–Feb 12, 2023	0.086	2.22	9.95	2.22	10.8	25.3	150
Feb 13, 2023–Feb 14, 2023	0.080	2.12	9.45	1.97	10.3	23.9	108
Feb 18, 2023–Feb 19, 2023	0.077	2.08	9.35	1.96	10.4	23.9	129
Feb 21, 2023–Feb 22, 2023	0.096	2.62	9.55	1.97	10.6	24.9	211
Feb 24, 2023–Feb 25, 2023	0.071	1.62	9.05	1.62	9.02	21.4	107
Feb 26, 2023–Feb 27, 2023	0.090	2.32	9.75	2.67	10.6	25.4	114
Min	0.071	1.62	9.05	1.62	9.02		
Max	0.104	2.72	13.0	2.93	14.0		
Mean $\pm$ SD ^a ($\mu\text{g}/\text{m}^3$)	0.086 $\pm$ 0.010	2.27 $\pm$ 0.332	10.2 $\pm$ 1.29	2.23 $\pm$ 0.424	11.0 $\pm$ 1.45	25.8 $\pm$ 3.38	155 $\pm$ 45.9

^aSD: standard deviation.

TABLE 2: Comparison of concentrations ($\mu\text{g}/\text{m}^3$) of water-soluble inorganic anions in $\text{PM}_{2.5}$ nationally and globally.

Country/location	Period	Fluoride	Chloride	Nitrate	Phosphate	Sulfate	References
Bangladesh, Dhaka	Feb 2, 2023–Feb 23, 2023	0.086 $\pm$ 0.010	2.27 $\pm$ 0.332	10.2 $\pm$ 1.29	2.23 $\pm$ 0.424	11.0 $\pm$ 1.45	Present study
Bangladesh, Dhaka	March 2001–April 2001	0.13–0.35	1.23–3.08	3.23–3.86	0.08–0.16	9.11–13.2	[22]
Bangladesh, Dhaka	Jan 2013–Feb 2013	—	1.19 $\pm$ 2.03	7.75 $\pm$ 6.51	—	5.30	[27]
Taiwan, Kaohsiung	Nov 1998–April 1999	—	3.07	12.8	—	17.8	[28]
India, Chhattisgarh	July 2009–June 2010	—	2.06	3.16	—	6.75	[29]
China, Nanchang	June 2009	—	1.62	2.75	—	28.9	[30]
Spain, Alicante	Sept 2005–Aug 2006	—	0.52	1.53	—	4.20	[31]
Japan, Yokohama	Sept 2007–Aug 2008	—	0.21	0.96	—	3.80	[32]
Singapore	Jan 2000–Dec 2000	—	0.60	0.90	—	5.00	[33]
Switzerland, Bern	April 1998–March 1999	—	0.102	3.00	—	2.80	[34]

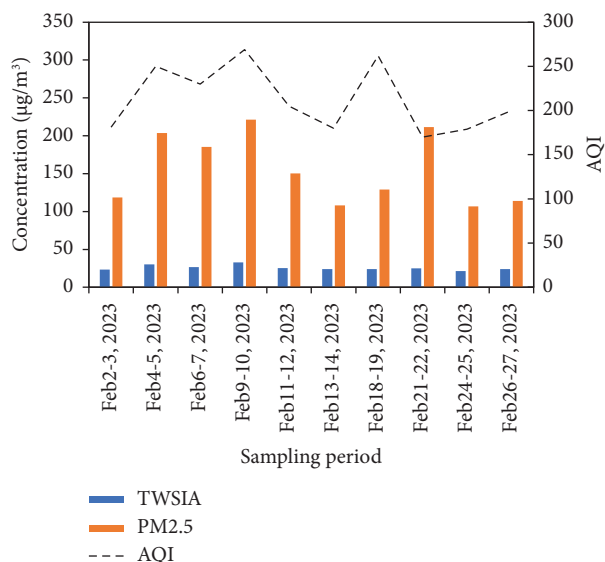


FIGURE 3: Daily variation of average mass concentration of PM_{2.5}, water-soluble inorganic anions, and AQI values obtained from global network [7].

chloride concentration. Chloride concentrations in PM_{2.5} are increased by recent automobile and construction activities as well. Given the high AQI values recorded in Dhaka since 2017 (Figure 2), the enhanced concentration of anions in PM_{2.5}, particularly chloride, is reasonable. In Taiwan, Lin found 3.07 µg/m³ of chloride in PM_{2.5} [28]. According to Deshmukh et al., the total concentration of chloride in India's Chhattisgarh was reported to be 2.06 µg/m³ (Table 2) [29]. Despite substantial urbanization and industry in China over the past few decades, it was found that the chloride concentration was only 1.62 µg/m³ [31]. The sampling was carried out in Nanchang, China, a developing city, but Beijing or Shanghai is predicted to have higher chloride concentrations. Reasonably, chloride concentrations in PM_{2.5} were substantially lower in Japan, Switzerland, Singapore, and Spain than in the current study (Table 2).

3.2.3. Nitrate. Nitrate concentrations were found to vary from 9 to 13 µg/m³, with the mean value being 10.2 ± 1.29 µg/m³ (Table 1). Similar to fluoride and chloride, nitrate was found at its highest (13 µg/m³) and lowest (9 µg/m³) concentrations during February 9-10, 2023 and February 24-25, 2023, respectively. A substantial amount of nitrate was found in PM_{2.5} in Dhaka due to the large number of vehicles, many of which are unfit, that are on the streets and generate high levels of NO_x into the atmosphere. Nitrate concentrations in PM_{2.5} in Dhaka were found to be only 3.23–3.86 µg/m³ during March-April 2001 [22]; however, it was quite high (7.75 ± 6.51 µg/m³) during January-February 2013 [27].

As previously mentioned, the air quality in Dhaka has drastically declined in the past decade, which is why substantial concentrations of anions, particularly nitrate, have been detected in PM_{2.5}. It is unusual to find high nitrate concentrations in PM_{2.5} in Taiwan. In Kaohsiung, Taiwan,

Lin found 12.8 µg/m³ nitrate in PM_{2.5} (Table 2) [28]. Although there are more automobiles in Singapore and Japan than in Bangladesh, the concentration of nitrate there was determined to be quite low at 0.90 and 0.96 µg/m³, respectively (Table 2). These findings imply that the air pollution in Bangladesh, particularly Dhaka, is primarily caused by the operation of unfit vehicles [5, 6]. Taiwan is an advanced country; therefore, it makes sense that unfit vehicles would not be on the roads there. The high concentration of nitrate (12.8 µg/m³) detected in PM_{2.5} in Taiwan should come from multiple sources. Nitrate concentrations in PM_{2.5} were reported to be 3.16 and 2.75 µg/m³ in India and China, respectively (Table 3) [29, 30]. These results suggest that the sampling locations did not resemble Dhaka, a megacity, or the use-unfit vehicle bans in China and India. Reasonably, nitrate concentration was found to be minimum in Spain, Japan, and Singapore and these were 1.53, 0.96, and 0.90 µg/m³, respectively (Table 2) [31–33]. On the other hand, it is unexpected that the concentration of nitrate in PM_{2.5} in Switzerland was found to be as high as 3.00 µg/m³ (Table 2) [33].

4. Phosphate

Phosphate concentrations ranged from 1.62 to 2.93 µg/m³, with a mean value of 2.23 ± 0.42 µg/m³ (Table 2). During February 9-10, 2023, and February 24-25, 2023, the highest and lowest values were reported to be 2.93 and 1.62 µg/m³, respectively. Reasonably, phosphate concentrations were found to be low during March-April 2001 and January-February 2013 as reported by Salam and coworkers, and the values were 0.08–0.16 and 0.62 ± 1.00 µg/m³, respectively (Table 2) [22, 27].

The following are some potential sources of phosphate in PM_{2.5} in Dhaka: (i) due to rapid urbanization, there has been a considerable increase in the usage of fire retardants in a variety of matrices and (ii) the use of phosphate-based surfactants in the textile industry. Many attempts have been made over the past few decades to modernize Dhaka; therefore, many fire retardants are being used in interior work. Since the textile industry in Bangladesh is primarily focused on exports, Dhaka is home to hundreds of textile companies that process textiles using considerable amounts of phosphate-based surfactants. In fact, the waste management system is inadequate in Bangladesh, which causes ongoing severe environmental contamination.

4.1. Sulfate. In PM_{2.5}, sulfate concentrations ranged from 9 to 14 µg/m³, with a mean concentration of 11 ± 1.4 µg/m³ (Table 1). The maximum concentrations of sulfate (14 µg/m³) and the lowest (9 µg/m³) were found during February 9-10, 2023 and February 24-25, 2023, respectively, similar to what was found for other anions. In a previous study, during March-April 2001, sulfate concentration was found to be 10.1–13.2 µg/m³ [22], but that was reported as only 5.30 ± 2.07 µg/m³ during January-February 2013 (Table 2) [26].

Huang et al. reported a very high concentration of sulfate in PM_{2.5} in Nanchang, China, that was found to be 28.9 µg/m³ (Table 2) [30]. Despite being a developed nation, Taiwan

TABLE 3: Correlation matrix of the WSIA in PM_{2.5} with $r^2 > 0.800$.

	Fluoride	Chloride	Nitrate	Phosphate	Sulfate	PM _{2.5}	AQI
Fluoride	1						
Chloride	0.958 ²	1					
Nitrate	0.801 ¹	0.746	1				
Phosphate	0.822 ¹	0.778	0.859 ¹	1			
Sulfate	0.845 ¹	0.837 ¹	0.978 ²	0.881 ¹	1		
PM _{2.5}	0.846 ¹	0.860 ¹	0.761	0.574	0.800 ¹	1	
AQI	0.417	0.440	0.721	0.637	0.743	0.469	1

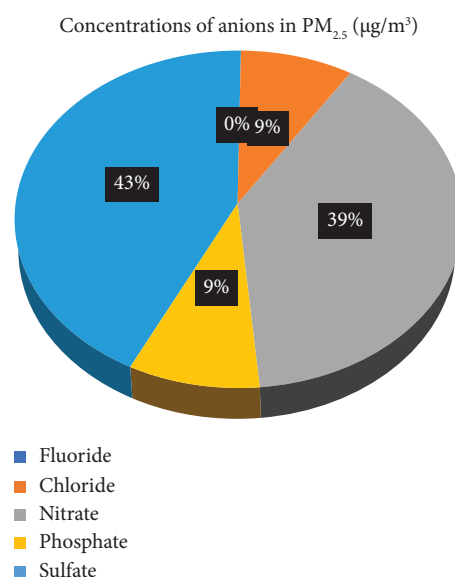
¹Correlation is significant at the 0.05 level (2-tailed).²Correlation is significant at the 0.01 level (2-tailed).

has a high concentration of sulfate in PM_{2.5}, which was measured at 17.8 $\mu\text{g}/\text{m}^3$ (Table 2) [31]. Sulfate concentration in India was reported to be 6.75 $\mu\text{g}/\text{m}^3$. Concentrations of sulfate in PM_{2.5} were logically lower in Spain, Japan, Singapore, and Switzerland than they were in Bangladesh, China, Taiwan, and India (Table 2) [22, 27–31, 38, 39]. Switzerland had the lowest sulfate concentration, i.e., 2.80 $\mu\text{g}/\text{m}^3$, whereas Japan, Spain, and Singapore had the highest concentrations, 3.80, 4.20, and 5.00 $\mu\text{g}/\text{m}^3$, respectively.

The usage of coal to heat homes and workplaces during the winter may be the cause of the high sulfate concentrations in China and Taiwan. The presence of elemental sulfur in diesel also results in SO_x emissions into the atmosphere, which raises the sulfate concentration of PM_{2.5}. However, unlike China or India, Bangladesh does not use coal in the same way. Brickfields and a small number of thermal power plants have historically used coal as a source of sulfate, but the high concentration of sulfate in PM_{2.5} points to an additional source of sulfate in Dhaka. The typical sulfur content of diesel in Bangladesh is around 350 ppm; however, recently, Bangladesh Petroleum Corporation imported diesel with a maximum sulfur content of 2500 ppm and blended it 20% with diesel that has 50 or 100 ppm sulfur [39]. However, India only allows 10 ppm of sulfur in diesel at most. Therefore, it might be inferred that utilizing diesel in Dhaka with a high sulfur content is the main source of sulfate in PM_{2.5}. The combustion of coal in the brickfields that surround Dhaka city also produces SO_x, which raises the concentration of sulfate in PM_{2.5} as well [4].

According to Zhang et al., the mass concentration ratio of nitrate to sulfate is frequently employed to estimate the contributions of stationary (coal fired) and mobile (vehicle exhaust) sources to the mass concentration of particulate matter [38]. When the ratio of nitrate/sulfate is greater than 1, it means that mobile sources are more responsible than fixed sources for the mass concentration of PM_{2.5}, whereas an inverse ratio has the opposite effect.

Figure 4 demonstrates that the percentages of nitrate and sulfate concentrations are almost equal, giving their ratios a close approximation to 1. This is consistent with the idea that both types of sources contribute equally to the total mass of nitrate and sulfate in PM_{2.5}. Numerous brickfields in the vicinity of Dhaka, as previously mentioned, use coal as

FIGURE 4: Water-soluble inorganic ions in PM_{2.5} during the sampling period.

their primary fuel source. Because coal contains considerable amounts of sulfur, the concentration of sulfate in PM_{2.5} in January and February is higher.

4.2. Statistical Analysis

4.2.1. Correlation Coefficient. Table 3 shows the correlation coefficients of fluoride, chloride, nitrate, sulfate, phosphate, PM_{2.5}, and AQI. The correlation coefficients for the WSIA, PM_{2.5}, and AQI were used to identify their common sources. The coefficients larger than 0.800 are highlighted in bold in Table 3. As can be seen in Table 3, there was a very substantial correlation of fluoride with chloride ($r^2 = 0.958$) and good correlation with nitrate ($r^2 = 0.801$), phosphate ($r^2 = 0.822$), sulfate ($r^2 = 0.845$), and PM_{2.5} ($r^2 = 0.846$). Phosphate also exhibited correlation with nitrate ($r^2 = 0.859$). Sulfate and nitrate exhibited a highly notable correlation ($r^2 = 0.978$), pointing to fossil fuel combustion from automobiles as one of their probable common origins [9, 22]. Sulfate also exhibited correlation with chloride ($r^2 = 0.837$) and phosphate ($r^2 = 0.881$). The correlation coefficients between PM_{2.5} and nitrate and sulfate, which are

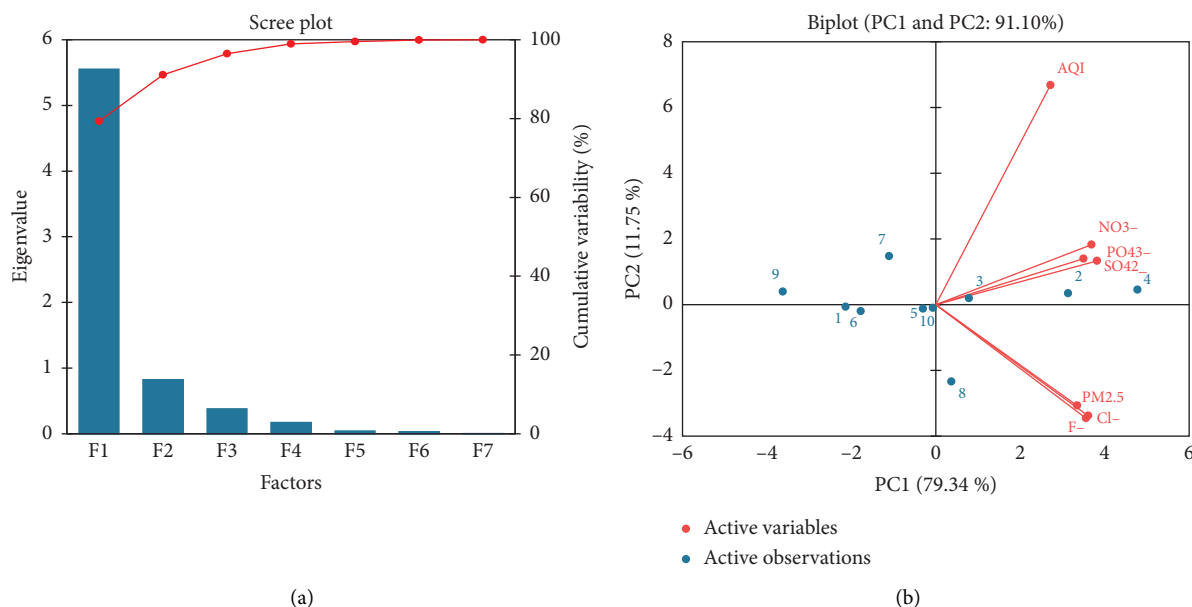


FIGURE 5: PCA (a) scree plot and (b) biplot of the first two components for source apportionment.

$r^2 = 0.761$ and $r^2 = 0.799$, respectively, suggested that they might share the same source. As mentioned above, sulfate and nitrate have an excellent correlation ($r^2 = 0.978$), which suggested that their sources may be identical. This is due to the high sulfur content of diesel fuel in Bangladesh and the use of unfit automobiles in Dhaka. Unfit automobiles release NO_x into the environment, whilst diesel with sulfur releases SO_x . These findings suggested that sulfate and nitrate concentrations in $\text{PM}_{2.5}$ are rising. The results revealed that fluoride and chloride originated from common sources, while nitrate, sulfate and phosphate come from different common sources. Vehicles using gasoline, diesel, natural gas, or coal combustion may release nitrate, sulfate, and phosphate, and on the other hand, fire retardants, burning of textile waste, and municipal solid waste may produce fluoride and chloride [22, 27]. It should be noted that all of the sources listed above can be considered the possible sources of $\text{PM}_{2.5}$ [22, 27].

4.2.2. PCA. The PCA results were used to assess the correlation among the variables such as fluoride, chloride, nitrate, sulfate, phosphate, $\text{PM}_{2.5}$, and AQI. Figures 5(a) and 5(b) show the PCA scree plot and biplot for the WSIA, $\text{PM}_{2.5}$, and AQI, respectively. The scree plot is a graphical tool that shows the variance explained by each component in PCA. It plots the eigenvalues, which are the measures of how much each component contributes to the total variance, against the component number. As shown in Figure 5(a), PC1 contributed 5.55 eigenvalues out of a total of 7.00, while PC2 provided the second-highest eigenvalue, 0.82. Thus, the other principal components do not significantly contribute to correlation.

A PCA biplot is an enhanced scatterplot that uses both points and vectors to represent structure. A biplot uses points to represent the scores of the observations on the

principal components, and it uses vectors to represent the coefficients of the variables on the principal components. Points that are close together correspond to observations that have similar scores on the components displayed in the plot. To the extent that these components fit the data well, the points also correspond to observations that have similar values on the variables. Figure 5(b) shows the PCA biplot for the two components which assess the correlation among the AQI, $\text{PM}_{2.5}$, and WSIA in order to identify possible sources. As can be seen in Figure 5(b), all of the variables have vector lengths that are almost equal, with the exception of the AQI variable with the longest line. The AQI variable has the largest variance of values than other variables. Nitrate, sulfate, and phosphate variables have a strong positive correlation because they have very small degree between three lines. Fluoride, chloride, and $\text{PM}_{2.5}$ vector lengths all showed a strong positive correlation since their degrees of vector lengths are so small. Negative correlations were found between fluoride/chloride/ $\text{PM}_{2.5}$ and nitrate/sulfate/phosphate and also between AQI and fluoride/chloride/ $\text{PM}_{2.5}$ because of high degrees of the vector lines. No correlation was observed between AQI and nitrate/sulfate/phosphate because the angle between the vector lines is more than 90° .

The PC1 and PC2 explained the highest variability, at 79.34% and 11.75%, respectively, toward the total variability (Supporting Figure S2). Comparatively, PC1 was maximum positively correlated (79.34%) with nitrate, sulfate, phosphate, and AQI but negatively correlated with fluoride, chloride, and $\text{PM}_{2.5}$, with excluding AQI (Supporting Figure S2).

The PCA was also performed to figure out the correlation between the remaining variables when the AQI and $\text{PM}_{2.5}$ variables were excluded, as indicated in Supporting Figures S3 and S4, respectively. Although no significant change was observed in PCA scree plots (Supporting Figures S3 and S4), variation in biplots was observed. As

observed before, positive correlations were found for the groups of fluoride/chloride/PM_{2.5} and nitrate/sulfate/phosphate excluding AQI; however, no correlation was found for PM_{2.5}/nitrate and PM_{2.5}/phosphate because the angle is either 90° and/or more than 90°. In variability, PC1 and PC2 contributed 85.36% and 8.36%, respectively. Excluding both AQI and PM_{2.5}, positive correlations were found for the groups of fluoride/chloride and nitrate/sulfate and phosphate, and no correlation was observed for fluoride/chloride with nitrate because of either and/or more than 90° angle between the vector lines (Supporting Figure S4).

5. Conclusion

The worst air quality levels were recorded in Dhaka over the years. Most days between January and February ranged from unhealthy to very unhealthy. The average PM_{2.5} mass was found to be $156 \pm 44.8 \mu\text{g}/\text{m}^3$, which was over three and a half times higher than that of the Bangladesh Department of Environment ($65 \mu\text{g}/\text{m}^3$) and about 10 times higher than that recommended by the WHO ($15 \mu\text{g}/\text{m}^3$). The results from this study were slightly higher than those from previous studies. Among the relevant anions in PM_{2.5}, sulfate constituted the majority of the anions altogether. Chloride, nitrate, and sulfate concentrations were three times greater than those previously reported in Dhaka. These findings and the AQI values suggest that the air quality in Dhaka has significantly deteriorated. The results of Pearson's correlation coefficients and PCA suggested that fossil fuels, such as sulfur-containing diesel, coal, and unfit vehicles, release high levels of SO_x, NO_x, and elemental carbon into the atmosphere, which causes an increase in the concentrations of nitrate and sulfate in PM_{2.5}. The open combustion of municipal solid wastes that contain bleaching powder may be one of the primary sources of chloride. Fluoride in PM_{2.5} indicates that fluorinated alkyl substances are extensively used in the production of rubber, plastic, textile components, etc. High levels of PM_{2.5} emissions are caused by extensive construction, industrialization, and the operation of unfit vehicles in Dhaka.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Ethics Statement

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

Riadul Rumman: formal analysis, data curation, and methodology; Md Rezaul Haq: data curation and methodology; Md Masudur Rahman Rahat: formal analysis, data curation, and methodology; Farhana Jahan: methodology; Shyama Prosad Moulick: methodology; Refayat Nigar: data

curation and methodology; Chandan Halder: data curation and methodology; Wenbin Liu: conceptualization and supervision; Farhana Khanam Ferdousi: supervision, fund acquisition, and final reviewing and editing; Ahsan Habib: conceptualization, formal analysis, data curation, investigation, methodology, supervision, fund acquisition, and writing original draft–review and final editing.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. (*Supporting Information*)

Supporting Table S1 indicates sample description along with their volumes of filtered air and daily average concentration of PM_{2.5} ($\mu\text{g}/\text{m}^3$). Table S1 shows that the standard volume of filtered air ranged from 132 to 140 m³, with a fixed period of 1440 min. The 24-h average PM_{2.5} mass was found to be $156 \pm 44.8 \mu\text{g}/\text{m}^3$ (day and night) during the study period. The value was almost two and half times higher than the Bangladesh Department of Environment guidelines of $65.0 \mu\text{g}/\text{m}^3$ [27] and nearly 10 times higher than the WHO guideline values of $15 \mu\text{g}/\text{m}^3$ [10]. The mean value of PM_{2.5} was found to be $191 \mu\text{g}/\text{m}^3$ in the samples collected between January 2013 and February 2013 (Table 2) [27].

During February 2, 2023, to February 27, 2023, an active air sampler (Model: TH-150H, China) with a 90 mm quartz filter (PALL Company, USA) was used at a flow rate of 70 L/min to collect PM_{2.5} for the purpose of analyzing WSIA. Supporting Figure S1 shows the photographs of the quartz filters before (a) and after (b) the filtration. Each sampling session lasted for 24 h (from 8:00 a.m. to the following 8:00 a.m.). The quartz filter was divided into four pieces and only a portion of it was treated in 20 mL distilled water using an automated shaker and sonication for 30 min. A detail description of sample collection and chemical analysis has been included in Sections 2.2 and 2.3 of the text.

The PCA variable plot can be used to assess the correlation among the variables. Loading plots also hint at how variables correlate with one another: a small angle implies positive correlation, a large one suggests negative correlation, and a 90° angle indicates no correlation between two characteristics. Supporting Figure S2 indicates variable plot for assessing the correlation among the WSIA, PM_{2.5}, and AQI.

PCA has been applied for the source apportionment of the WSIA, PM_{2.5}, and AQI. Supporting Figure S3 indicates scree plot, biplot, and variable plot in PCA for assessing correlation among the WSIA and PM_{2.5}.

The source apportionment of WSIA, PM_{2.5}, and AQI has been accomplished via PCA. Supporting Figure S4 indicates scree plot, biplot, and variable plot in PCA to assess correlation among the WSIA.

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